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Education

- 2023–2027 **Ph.D., Computer Science**, Johns Hopkins University, Baltimore, MD
- Member of the **Biomechanical- and Image-Guided Surgical Systems (BIGSS) Lab**, Johns Hopkins University.
 - Research Assistant at the **Institute for Integrative and Innovative Research (I³R)**, University of Arkansas.
 - Research focuses on **medical robotics, medical image analysis, computer vision, image-guided intervention, and augmented reality**.
 - *Expected graduation: December 2027.*
- 2021–2022 **M.S.E., Robotics**, Johns Hopkins University, Baltimore, MD
- Member of the **BIGSS Lab**.
 - Research focused on **medical robotics and image-guided surgical systems**.
- 2016–2020 **B.Eng., Mechanical Engineering**, University of Leeds, Leeds, UK
- Research concentration in **imitation learning and autonomous driving**.
 - Developed learning-based methods for **self-driving vehicle control and decision-making**.

Research Experience

- 2023–Present **Graduate Research Assistant**, Johns Hopkins University, Baltimore, MD
- **Real-Time Robotic Perception & Multimodal Sensing:** Architected a TensorRT-accelerated CNN–Transformer pipeline for fluoroscopic segmentation, fused with optical tracking and robot encoder data via ROS 2 and NVIDIA Isaac ROS; achieved **0.68 mm 3D reconstruction error** at **47 ms end-to-end latency**.
 - **C-Arm Kinematics & Positioning Guidance:** Developed and validated 6-DoF positioning guidance integrating geometric calibration, forward/inverse kinematics, optical tracking, and real-time pose estimation; **reduced X-ray usage by 87.2%** and **perceived workload by 72.8%** in a 12-participant user study.
 - **Fluoroscopic AR Navigation:** Led development of a real-time image-guided system integrating C-arm calibration, HoloLens tracking, CUDA-accelerated DRR generation, X-ray augmentation, and ZeroMQ communication; **reduced X-ray usage by 62.3%** and increased breach-free screw diameter from **5.9 mm to 12.95 mm** in surgeon evaluation.
 - **MRI-to-Fluoroscopy Registration:** Co-led an end-to-end framework combining U-Net segmentation, cGAN-based MRI-to-CT synthesis, DRR generation, and multi-view 2D–3D registration; achieved **2.4 ± 1.0 mm** translation and **1.6 ± 0.8°** rotation error in cadaveric evaluation.

tion. [\[Paper\]](#)

- **MRI Segmentation & Osteonecrosis Localization:** Developed 2D/3D U-Net models for T1-weighted MRI segmentation of the femur, pelvis, and osteonecrotic lesions, achieving **0.93 Dice** for femur and **0.91 Dice** for pelvis; integrated segmentation, registration, and DRRs for intraoperative localization of pathology not visible under fluoroscopy. [\[Paper\]](#)
- **Multi-Device Tracking & Pose Optimization:** Developed a device-agnostic framework integrating HoloLens 2 and optical trackers through ZeroMQ, dynamic scene graphs, and pose-graph optimization; reduced translational error by approximately **80%** and tracking-loss rate by **30%** versus the best single-device baseline.
- **Multimodal Pelvic Fracture AI Benchmark:** Built datasets and evaluation infrastructure for CT/X-ray pelvic fracture segmentation and co-organized MICCAI PENGWIN challenges, spanning **48,600 simulated X-rays** and later expanding to **500 CT cases and 340 participants worldwide**; 2024 benchmark results published in *IEEE TMI*. [\[Paper\]](#)

2026–Present **Research Assistant**, I³R, University of Arkansas, Fayetteville, AR

- **X-ray Depth Estimation:** Developing a DINO–DPT single-view depth network; built **11,000 synthetic X-ray/depth pairs** using CUDA-accelerated DRR generation and ray casting, plus **500+ real X-rays** with CT-derived depth recovered through 2D–3D registration.
- **CT-to-Fluoroscopy Navigation & Localization:** Leading development of a 2D–3D pipeline registering preoperative CT segmentations and surgical plans to intraoperative fluoroscopy, integrating TensorRT-accelerated perception and multimodal sensor fusion for real-time anatomical localization and trajectory guidance.
- **Continuum-Robot Teleoperation:** Developed and evaluated image-guided teleoperation for femoral-head core decompression, integrating preoperative planning, real-time manipulator perception, visualization, and steerable continuum-robot manipulation for cadaveric drilling.
- **Robot Control & System Integration:** Developing a control stack combining UR10e Cartesian and force control, custom continuum-robot actuation, force–torque sensing, multimodal perception, and steerable-manipulator control toward automated execution of registered surgical plans.

Other Experience

2022–2023 **Computer Vision Engineer**, PediaMetrix, Rockville, MD

- **RGB-D 3D Reconstruction:** Developed an end-to-end infant-head reconstruction pipeline combining deep-learning segmentation, point-cloud generation, multi-frame registration, bundle adjustment, and geometric processing for robust reconstruction under motion and depth artifacts.
- **3D Cranial Morphometrics:** Designed automated geometric measurement algorithms using camera geometry, facial landmarks, smoothing, and outlier rejection; achieved **5.0 mm MAE (1.1%)** in head circumference and **1.6-point MAE (1.9%)** in cranial index.
- **Clinical CV/ML Validation:** Validated and refined a 2D cranial-shape measurement pipeline across **445 cases**, achieving **0.94–0.95 Spearman correlation** with 3D reference measurements; results presented at Pediatric Academic Societies 2023.

Publications

 Google Scholar

† → Equal contribution

Journal Articles

- J1. Sang, Y. *et al.* Benchmark of Segmentation Techniques for Pelvic Fracture in CT and X-Ray: Summary of the PENGWIN 2024 Challenge. *IEEE Transactions on Medical Imaging* **45**, 2212–2228. <https://doi.org/10.1109/TMI.2025.3650126> (2026).
- J2. Zou, X., Liu, M., Jones, T. T., Millan, B., Gurram, S., Pinto, P. A., Freidlin, R. & Martin-Gomez, A. Visual Cue Interactions in AR-Guided Needle Insertion: A Prostate Biopsy-Inspired Phantom Study. *IEEE Transactions on Visualization and Computer Graphics*. arXiv: 2608.27620. <https://arxiv.org/abs/2608.27620> (2026).
- J3. Ku, P.-C., Liu, M., Grupp, R., Harris, A., Oni, J. K., Mears, S. C., Martin-Gomez, A. & Armand, M. End-to-End 2D/3D Registration from Pre-Operative MRI to Intra-Operative Fluoroscopy for Orthopedic Procedures. *International Journal of Computer Assisted Radiology and Surgery* **20**. Ping-Cheng Ku and Mingxu Liu contributed equally., 2355–2366. <https://doi.org/10.1007/s11548-025-03426-w> (2025).
- J4. Shu, H., Liu, M., Seenivasan, L., Gu, S., Ku, P.-C., Knopf, J., Taylor, R. H. & Unberath, M. Seamless Augmented Reality Integration in Arthroscopy: A Pipeline for Articular Reconstruction and Guidance. *Healthcare Technology Letters* **12**, e12119. <https://doi.org/10.1049/ht12.12119> (2025).
- J5. Zhang, H., Killeen, B. D., Ku, Y.-C., Seenivasan, L., Zhao, Y., Liu, M., Yang, Y., Gu, S., Martin-Gomez, A., Taylor, R. H., Osgood, G. & Unberath, M. StraightTrack: Towards Mixed Reality Navigation System for Percutaneous K-Wire Insertion. *Healthcare Technology Letters* **11**, 355–364. <https://doi.org/10.1049/ht12.12103> (2024).
- J6. Liu, M., Martin-Gomez, A., Oni, J. K., Mears, S. C. & Armand, M. Towards Visualizing Early-Stage Osteonecrosis Using Intraoperative Imaging Modalities. *Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization* **11**, 1234–1242. <https://doi.org/10.1080/21681163.2022.2157329> (2023).

Peer-reviewed Conference Proceedings

- C1. Zhang, J., Liu, M., Shu, H., Liang, R., Liu, Y., Taskar, O., Kheradmand, A., Armand, M. & Martin-Gomez, A. Extend Your Horizon: A Device-Agnostic Surgical Tool Tracking Framework with Multi-View Optimization for Augmented Reality. In *2026 IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)* 678–685 (IEEE, 2026). <https://doi.org/10.1109/VR67842.2026.00087>.
- C2. Zou, X., Liu, M., Shu, H., Liang, R., Unberath, M. & Martin-Gomez, A. The Role of Mixed and Augmented Reality in Medical Visualization: Literature Review and A Context-Aware Taxonomy. In *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* Accepted (2026). arXiv: 2608.27644. <https://arxiv.org/abs/2608.27644>.

Manuscripts Under Review

- W1. Liu, M., Liu, Z., Cai, R., Ku, Y.-C., Gu, S., Jain, A., Martin-Gomez, A. & Armand, M. *X-GuideAR: An Augmented Reality Framework to Mitigate Radiation Exposure during Fluoroscopic Guidance* Manuscript under review at International Journal of Computer Assisted Radiology and Surgery. 2026. arXiv: [2607.10873](https://arxiv.org/abs/2607.10873) [cs.CV]. <https://arxiv.org/abs/2607.10873>.
- W2. Shu, H., Seenivasan, L., Liu, M., Hwang, Y., Ku, Y.-C., Knopf, J., Martin-Gomez, A., Armand, M. & Unberath, M. *DualVision ArthroNav: Investigating Opportunities to Enhance Localization and Reconstruction in Image-Based Arthroscopy Navigation via External Cameras* Preprint. 2025. arXiv: [2511.10699](https://arxiv.org/abs/2511.10699) [cs.CV]. <https://arxiv.org/abs/2511.10699>.

Academic Service

2026	Organizer , PENGWIN 2026, MICCAI 2026	Apr. 2026–Oct. 2026
2026	Reviewer , IEEE International Symposium on Mixed and Augmented Reality (ISMAR)	
2024	Organizer , PENGWIN 2024, MICCAI 2024	Apr. 2024–Oct. 2024

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